# Employee Salary and Department Analysis — Summary of Findings

## 1. Problem Statement

The objective of this analysis is to understand employee distribution across departments and cities, examine salary patterns, and identify useful insights to support workforce planning and salary-related decisions.

## 2. Data Description

The dataset contains **8 employees and 5 columns**: employee ID, name, department, salary, and city. It includes four departments—Sales, HR, IT, and Finance—and three cities—Chennai, Bangalore, and Mumbai. The analysis uses summary statistics, department and city comparisons, and visualizations such as bar charts, histograms, line charts, scatter plots, and box plots.

## 3. Three Key Insights

**Insight 1: IT has the highest employee count.**
The IT department has 3 employees, followed by HR and Sales with 2 employees each, while Finance has 1 employee. IT accounts for 37.5% of the employees in this dataset.

**Insight 2: Salaries vary considerably.**
The average monthly salary is ₹58,500, while the median is ₹56,000. Salaries range from ₹30,000 to ₹90,000, resulting in a difference of ₹60,000 between the minimum and maximum salaries.

**Insight 3: Mumbai has the highest average salary by city.**
Mumbai has an average monthly salary of ₹69,000, followed by Bangalore at ₹56,000 and Chennai at ₹54,500. However, Mumbai and Bangalore each have only two employees, so these differences should be interpreted cautiously.

## 4. Recommendations

**Recommendation 1: Conduct a detailed salary review.**
Compare salaries by department, job role, experience, and seniority to understand pay differences and identify potential inconsistencies. Avoid making salary decisions based only on department or city averages.

**Recommendation 2: Collect more comprehensive employee data.**
Expand the dataset with additional employee records and relevant fields such as job title,

years of experience, joining date, and performance rating. This would support more reliable analysis and better workforce planning.

## 5. Limitations

The analysis is based on only 8 employees and may not represent the entire organization. The dataset does not contain job titles, experience, performance ratings, or historical salary information. Therefore, the line chart displays sorted salary values rather than a time trend. Employee ID is an identifier and does not provide a meaningful business relationship with salary. No outliers were identified using the IQR rule, but this does not guarantee that every salary value is correct. The findings describe this sample and should not be generalized without further data.

## Conclusion

The analysis identifies differences in employee distribution and salary levels across departments and cities. These findings provide an initial overview of the workforce, but a larger and more detailed dataset is needed before making important business decisions.